National University of Computer and Emerging Sciences
Lahore Campus

[redacted]

Engineering Economics (MG3008)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 55
Total Questions: 3

Date: February 27 2025

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

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Attempt all the questions.

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q1: Describe the following.(Not more than few lines) [marks 15]

1) Opportunity Cost
2) Elasticity
3) Micro & Macro Economics

CLO # 2: Interpret the concept of time value of money

Q2: A Company's transactions for the last year are given below. [marks 20]

Year 1 through 5: Invest \$7000 in year 1, decreasing by \$1000 per year through year 5.
Year 6 through 10: No new investment and no withdrawals.
Year 11 through 15: Withdraw \$20,000 in year 11, decreasing 20% per year through year 15.
By using the time value of money, Determine the total present worth if the interest rate was 10 % per year.

CLO # 2: Interpret the concept of time value of money

Q3: For the cash flows shown in the diagram, Compute the value of x that will make the future worth in year 8 equals to \$-70,000. Use interest rate of 12% per year. [marks 20]